Summary

Sensing the environment Current

14 of 19 weeks

Subject	Year	Start date	Duration
Design	MYP2	Week 4, July	19 weeks

Description

The "Sensing the Environment" unit is designed to provide students with an opportunity to apply their electronics knowledge in the creation of a device that can interact with the surrounding environment. In this unit, students will explore various environmental sensors such as temperature, light, humidity, or motion sensors, and learn how to integrate them with electronic components.

Key and Related Concepts

Key Concepts

Key Concepts	Definition
Systems	**Systems** are sets of interacting or interdependent components. Systems provide structure and order in human, natural and built environments. Systems can be static or dynamic, simple or complex.

Related Concept(s)

Function

Inquiry

Conceptual Understanding

"Exploring how systems of sensors operate and contribute to various functions in our environment."

Global Context

Global Context	Explorations to develop
Scientific and technical innovation	Systems, Models

Statement of Inquiry

"Exploring how systems of sensors function within scientific and technical methods to sense and understand the environment."

Inquiry Questions

Type	Inquiry Questions	Line of Inquiry
Factual	What is a sensor and what does it do?	Sensors definition
Factual	Can you think of things around us that might use sensors to work better?	Environment
Conceptual	How do sensors help machines "understand" the world around them?	sensors, environment
Conceptual	What are some different ways that sensors can communicate information to us?	Different types of sensors
Debatable	Should sensors be used to monitor how people drive cars to improve road safety?	Application of sensors
Debatable	Should schools use sensors to keep track of where students are during the day?	Application of sensors

Curriculum

Aims

Develop knowledge, understanding and skills from different disciplines to design and create solutions to problems using the design cycle

Use and apply technology effectively as a means to access, process and communicate information, model and create solutions, and to solve problems

Develop an appreciation of the impact of design innovations for life, global society and environments

MYP subject group objective(s)

A: Inquiring and analysing

i. explain and justify the need for a solution to a problem

ii. construct a research plan, which states and prioritizes the primary and secondary research needed to develop a solution to the problem

iii. analyse a group of similar products that inspire a solution to the problem

iv. develop a design brief, which presents the analysis of relevant research

B: Developing ideas

i. develop a design specification, which outlines the success criteria for the design of a solution based on the data collected

ii. present a range of feasible design ideas, which can be correctly interpreted by others

iii. present the chosen design and outline the reasons for its selection

iv. develop accurate planning drawings/diagrams and outline requirements for the creation of the chosen solution

C: Creating the solution

i. construct a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution

ii. demonstrate excellent technical skills when making the solution

iii. follow the plan to create the solution, which functions as intended

iv. explain changes made to the chosen design and plan when making the solution

D: Evaluating

i. describe detailed and relevant testing methods, which generate accurate data, to measure the success of the solution

ii. explain the success of the solution against the design specification

iii. describe how the solution could be improved

iv. describe the impact of the solution on the client/target audience

Content (topics, knowledge, skills)

Factual Knowledge:

- Understanding what sensors are and how they detect and measure things in the environment.

- Knowing examples of sensors in everyday life and their functions.

- Recognizing that sensors contribute to making machines "smart" by providing information.

Conceptual Knowledge:

- Grasping that sensors are part of systems and contribute specific functions to those systems.

- Understanding that sensors help machines respond to changes in their environment.

- Recognizing that sensors play a role in scientific and technical innovations.

Procedural Knowledge:

- Knowing how to identify basic types of sensors and their uses.

- Understanding how to set up and use simple sensors.

- Gaining skills in interpreting sensor readings and understanding their significance.

Skills

Skills for Meeting Objectives and Summative Assessment:

Observation: Ability to watch and notice changes in the environment that sensors detect.

Classification: Skill to group different sensors based on their functions.

Application: Using sensors in simple setups to gather information and make conclusions.

Interpretation: Understanding what sensor readings mean and how they relate to the environment.

Problem-Solving: Solving challenges that arise while setting up or using sensors.

Communication: Explaining how sensors work and their role in different systems.

Critical Thinking: Evaluating when and how sensors are beneficial in various scenarios.

Innovation: Imagining new ways to use sensors for better understanding and control.

ATL Skills

ATL skills

Self-management

- V. Reflection skills

(Re-)considering the process of learning; choosing and using ATL skills

Develop new skills, techniques and strategies for effective learning

Identify strengths and weaknesses of personal learning strategies (self-assessment)

Try new ATL skills and evaluate their effectiveness

Consider content

What did I learn about today?

What don't I yet understand?

What questions do I have now?

Consider ATL skills development

What can I already do?

How can I share my skills to help peers who need more practice?

What will I work on next?

Consider personal learning strategies

What can I do to become a more efficient and effective learner?

How can I become more flexible in my choice of learning strategies?

What factors are important for helping me learn well?



Research

- VI. Information literacy skills

  Finding, interpreting, judging and creating information

    Collect, record and verify data

    Understand and use technology systems



Thinking

- VIII. Critical thinking skills

  Analysing and evaluating issues and ideas

    Use models and simulations to explore complex systems and issues

- X. Transfer skills

  Utilizing skills and knowledge in multiple contexts

    Combine knowledge, understanding and skills to create products or solutions

Developing IB Learners

IB Learner Profile



Inquirers



Knowledgeable



Thinkers



Communicators



Caring



Reflective

Description

Throughout the unit, I would consistently connect specific activities and tasks to different attributes of the Learner Profile. For instance:

- To promote "Inquirer" attributes, I would encourage students to explore various sensor technologies, research their

applications, and ask probing questions about their potential impact on the environment.

- To foster "Thinker" attributes, I would prompt students to critically evaluate different sensor integration approaches, considering both technical feasibility and ethical implications.

- For "Communicator" attributes, I would emphasize the importance of clear and concise documentation of design processes, enabling effective communication of their projects to peers and potential stakeholders.

- To encourage "Reflective" attributes, I would incorporate regular moments for students to pause and reflect on their progress, challenges faced, and lessons learned. This reflection could be done individually or in small groups, encouraging students to learn from their experiences.

Connections

Language Development

This unit will help students develop their language skills by encouraging them to describe, explain, and discuss concepts related to sensors and their functions. They will use vocabulary specific to sensors and technology, improving their ability to communicate effectively. Through group discussions and assignments, students will practice expressing their thoughts, ideas, and findings, fostering their language fluency and communication confidence.

Information Communication Technology

In this unit, students will learn how to find reliable information about sensors using digital tools like websites and online databases. They will practice evaluating sources to make sure the information is accurate and helpful. By using technology for research and presentations, they will develop their skills in using digital tools responsibly and effectively.

Service as Action

This unit engages your students in principled action by fostering their understanding of sensors' real-world applications and ethical considerations. To initiate service, we can encourage students to identify a problem in their community that sensors could help solve, such as environmental monitoring or safety enhancements. This unit supports learning outcomes related to critical thinking, problem-solving, ethical decision-making, and innovative thinking, as students explore the functions and impact of sensors within scientific and technical contexts.

Assessment

Formative Assessment

By below mentioned formative tasks we will be done to access the students' skills and understanding of the unit:

1. Sensors in Everyday Life Discussion: We will lead a class discussion where students brainstorm different scenarios where sensors are used in their daily lives. They can consider appliances, gadgets, and systems that rely on sensors. Each student shares their example, explains the sensor's function, and discusses how it improves convenience or safety.

2. Sensor Mind map (written): Students will be creating a mind map that explores different aspects of sensors, their functions, and their impact on the environment.

3. Sensor Exploration Stations: We shall set up stations around the classroom with different types of sensors (temperature sensors, light sensors, motion sensors, etc.). Each station should have a simple experiment for students to conduct, such as observing how sensor readings change in response to different conditions. Students rotate through the stations, record their observations, and discuss their findings afterward.

4. Sensor Simulation: Students will use a simulation software or app that mimics sensor behavior. They can adjust settings and observe how the sensor responds. This interactive approach helps them grasp sensor operation in a controlled environment.

OCT
15

Debug the System: ATL : Critical thinking: Analysing, Debugging, and Troubleshooting Systems

Formative Formative 16/18 Students Thursday at 10:05 AM

You are given a **faulty circuit + code** that is **not working as expected**. Your role is to act as a **troubleshooter**, analyse them carefully, identify errors, and fix them logically.

This task is **not about getting the right answer quickly**, but about **thinking critically**, testing ideas, and justifying decisions.

What will you do:

1. **Observe the Problem**
    a. Run the simulation as it is.
    b. Record what *should* happen vs what *actually* happens.
2. **Break the System into Parts**
    a. Identify inputs, outputs, and processes.
    b. Separate **circuit issues** from **software (code) issues**.
3. **Identify Possible Errors**
    a. List at least **3 possible reasons** why the system is not working.
    b. Categorise errors (circuits, logic, values, conditions).
4. **Test & Debug**
    a. Change **only one thing at a time**.
    b. Test after every change.
5. **Fix the System**
    a. Correct the identified errors.
    b. Ensure the simulation now works as intended.
6. **Justify Your Thinking**
    a. Explain:
        i. What the error was
        ii. Why it caused the problem
        iii. How your fix solved it

Task Specific Clarification

1–2

- Identify at least one error in the given circuit or code with guidance.
- Identify the input and output of the system.
- Make a simple correction and state what was changed.

3–4

- Describe what the circuit should do and what happens when the simulation runs.
- Identify at least two possible errors in the circuit or code.
- Make corrections and test whether the system works.
- Explain in simple words how the corrections helped.

5–6

- Identify the inputs, processes and outputs of the system.
- Identify at least three possible errors in the circuit or code and classify them as circuit or code errors.
- Change one thing at a time and test the simulation after each change.
- Correct the errors and explain why they caused the problem and how the fixes helped the system work as expected.

7–8

- Systematically investigate the circuit and code to identify and explain the causes of the errors.
- Identify at least three possible errors and distinguish between circuit connection errors, incorrect values and code logic errors.
- Follow a logical debugging process, changing one thing at a time and recording the results of each test.
- Correct the errors and test the system to confirm that it works as intended.
- Justify each correction by explaining the cause of the error, its effect on the system and why the solution works.
- Suggest one improvement that could make the circuit or code more reliable.

SEP
22

Isometric Sketching

Formative 13/18 Students Tuesday at 11:35 AM

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video1 video 2 vide o 3 Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the

Tissue_holder.png
1.39 MB

isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
1–2	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a basic sketch of the tissue box holder, but the design idea is difficult to interpret. Isometric lines, proportions and/or product features are frequently inaccurate or unclear.
3–4	Develop ideas that can be correctly interpreted by others.	The student produces an identifiable isometric sketch of the tissue box holder. Some isometric lines, proportions and visible features are represented correctly, although there are noticeable inaccuracies or inconsistencies.
5–6	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a clear and reasonably accurate isometric representation of the tissue box holder. Proportions, straight lines and visible features are mostly accurate, allowing the product to be understood clearly.
7–8	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a precise, well-proportioned and clearly communicated isometric sketch. Isometric lines are consistently used, visible features are accurately represented, and the sketch communicates the form of the tissue box holder effectively and confidently.

SEP 22

Isometric Sketching

Formative 17/18 Students Tuesday at 11:35 AM

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video1 video 2 vide o 3 Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the

Tissue_holder.png
1.39 MB

isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
1–2	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a basic sketch of the tissue box holder, but the design idea is difficult to interpret. Isometric lines, proportions and/or product features are frequently inaccurate or unclear.
3–4	Develop ideas that can be correctly interpreted by others.	The student produces an identifiable isometric sketch of the tissue box holder. Some isometric lines, proportions and visible features are represented correctly, although there are noticeable inaccuracies or inconsistencies.
5–6	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a clear and reasonably accurate isometric representation of the tissue box holder. Proportions, straight lines and visible features are mostly accurate, allowing the product to be understood clearly.
7–8	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a precise, well-proportioned and clearly communicated isometric sketch. Isometric lines are consistently used, visible features are accurately represented, and the sketch communicates the form of the tissue box holder effectively and confidently.

SEP
22

Isometric Sketching

Formative 17/20 Students Tuesday at 11:35 AM

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video1 video 2 vide o 3 Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the

Tissue_holder.png
1.39 MB

isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
1–2	Develop a range of	The student produces a basic sketch of the tissue box holder, but the design

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
	feasible design ideas that can be correctly interpreted by others.	idea is difficult to interpret. Isometric lines, proportions and/or product features are frequently inaccurate or unclear.
3–4	Develop ideas that can be correctly interpreted by others.	The student produces an identifiable isometric sketch of the tissue box holder. Some isometric lines, proportions and visible features are represented correctly, although there are noticeable inaccuracies or inconsistencies.
5–6	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a clear and reasonably accurate isometric representation of the tissue box holder. Proportions, straight lines and visible features are mostly accurate, allowing the product to be understood clearly.
7–8	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a precise, well-proportioned and clearly communicated isometric sketch. Isometric lines are consistently used, visible features are accurately represented, and the sketch communicates the form of the tissue box holder effectively and confidently.

SEP
22

Isometric Sketching

Formative 17/18 Students Tuesday at 11:35 AM

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video1 video 2 vide o 3 Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the

Tissue_holder.png
1.39 MB

isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
1–2	Develop a range of feasible design ideas	The student produces a basic sketch of the tissue box holder, but the design idea is difficult to interpret. Isometric lines, proportions and/or product features are

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
	that can be correctly interpreted by others.	frequently inaccurate or unclear.
3–4	Develop ideas that can be correctly interpreted by others.	The student produces an identifiable isometric sketch of the tissue box holder. Some isometric lines, proportions and visible features are represented correctly, although there are noticeable inaccuracies or inconsistencies.
5–6	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a clear and reasonably accurate isometric representation of the tissue box holder. Proportions, straight lines and visible features are mostly accurate, allowing the product to be understood clearly.
7–8	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a precise, well-proportioned and clearly communicated isometric sketch. Isometric lines are consistently used, visible features are accurately represented, and the sketch communicates the form of the tissue box holder effectively and confidently.

SEP
10

Isometric Sketching

Formative 12/18 Students Thursday at 10:05 AM

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video1 video 2 vide o 3 Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the

Tissue_holder.png
1.39 MB

isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
1–2	Develop a range of feasible design ideas that can be correctly	The student produces a basic sketch of the tissue box holder, but the design idea is difficult to interpret. Isometric lines, proportions and/or product features are frequently inaccurate or unclear.

Grade Band	IB Rubric – Criterion B ii	Task-Specific Clarification (TSC)
	interpreted by others.	
3–4	Develop ideas that can be correctly interpreted by others.	The student produces an identifiable isometric sketch of the tissue box holder. Some isometric lines, proportions and visible features are represented correctly, although there are noticeable inaccuracies or inconsistencies.
5–6	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a clear and reasonably accurate isometric representation of the tissue box holder. Proportions, straight lines and visible features are mostly accurate, allowing the product to be understood clearly.
7–8	Develop a range of feasible design ideas that can be correctly interpreted by others.	The student produces a precise, well-proportioned and clearly communicated isometric sketch. Isometric lines are consistently used, visible features are accurately represented, and the sketch communicates the form of the tissue box holder effectively and confidently.

AUG
13

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 18/20 Students Thursday at 10:05 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research • Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

AUG 11

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 17/18 Students Tuesday at 11:35 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."

- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research • Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

AUG 11

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 18/18 Students Tuesday at 11:35 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research • Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

AUG
11

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 17/18 Students Tuesday at 11:35 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research

	• Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

AUG 11

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 16/18 Students Tuesday at 11:35 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12

My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research • Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

Summative Assessment

GRASP task

Have students create an interactive presentation using digital tools (such as slideshows or videos) to explain the concept of sensors, their functions, and their applications in the environment. The presentation should engage the audience and include visual demonstrations.

DEC 10

Criteria C and D_SA 2

Summative Summative 0/18 Students Thursday at 10:05 AM

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.
- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the**

Solution. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.
3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram and a reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.
3–4	i. Outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains changes made to the chosen design and plan when making the solution.	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii. Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Describes a testing method, which is used to measure the success of the solution. ii. States the success of the solution.	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the impact of the solution on the client/target audience, with guidance.	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit, code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii. Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.
- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

DEC
10

Criteria C and D_SA 2

Summative Summative 0/18 Students Thursday at 10:05 AM

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.
- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the Solution**. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.

3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram and a reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.
3–4	i. Outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains changes made to the chosen design and plan when making the solution.	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii. Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Describes a testing method, which is used to measure the success of the solution. ii. States the success of the solution.	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit,

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the impact of the solution on the client/target audience, with guidance.	code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii. Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.
- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

DEC
10

Criteria C and D_SA 2

Summative Summative 0/20 Students Thursday at 10:05 AM

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.
- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the Solution**. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.
3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram and a reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
3–4	i. Outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains changes made to the chosen design and plan when making the solution.	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii. Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
1–2	i. Describes a testing method, which is used to measure the success of the solution. ii. States the success of the solution.	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the impact of the solution on the client/target audience, with guidance.	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit, code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii. Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.
- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

DEC
10

Criteria C and D_SA 2

Summative Summative 0/18 Students Thursday at 10:05 AM

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.
- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the Solution**. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.
3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram

and a reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.
3–4	i. Outlines each step in a plan that contains some details, resulting in peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains changes made to the chosen design and plan when making the solution.	Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Describes a testing method, which is used to measure the success of the solution. ii. States the success of the solution.	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit, code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	impact of the solution on the client/target audience, with guidance.	
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii. Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.
- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

DEC
10

Criteria C and D_SA 2

Summative Summative 0/18 Students Thursday at 10:05 AM

SA Template

ATL Skills

Thinking Skills – Critical Thinking

- Evaluate evidence, arguments, and existing products to justify a proposed design solution.
- Identify strengths, limitations, and opportunities for improvement based on testing evidence.

Research Skills – Information Literacy

- Collect, record, and verify data from primary and secondary sources.

- Interpret testing results to evaluate the effectiveness of the designed solution.

Self-Management Skills – Organisation

- Plan and manage the steps required to create the solution.
- Organise technical evidence, document changes, and complete the ePortfolio systematically.

Concepts

Key Concept: Communities

Related Concepts: Function, Innovation

Content

Planning, creating, testing, and evaluating a sensor-based automated system using a sensor, Arduino microcontroller, and output component. Students will develop their chosen design into a simulation using Wokwi, demonstrate technical skills, document changes made during creation, test the system against its design specifications, and propose evidence-based improvements.

Statement of Inquiry

Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation

You are a student design engineer developing a sensor-based automated system to address a real community problem. Having investigated the problem, analysed existing products, developed design specifications, and selected a final design, you must now plan and create your proposed solution.

Your task is to translate your chosen design into a working simulation using Wokwi. The system must use an appropriate sensor, an Arduino microcontroller, and an output component to detect a condition and trigger a suitable response.

Throughout the creation process, you must demonstrate your technical skills, document the steps involved in developing the solution, and record any changes made to your original design or plan. You must also provide evidence of your circuit construction, Arduino code, and debugging or modification process.

Once your solution has been created, you will test its functionality and collect evidence of its performance. You will evaluate how effectively the system meets your design specifications and addresses the identified community problem. Based on your findings, you will identify its strengths and limitations and propose justified modifications for future improvement.

You will document your work in the **MYP Design ePortfolio: Criteria C & D – Creating & Evaluating the Solution**. Your submission must demonstrate a clear connection between your chosen design, the solution you create, the evidence collected through testing, and your evaluation.

Role

A student design engineer responsible for planning, creating, testing, and evaluating a sensor-based automated solution for a real community problem.

Audience

The intended users of the proposed solution, members of the affected community, and the design panel evaluating

the functionality, effectiveness, and potential for improvement of the solution.

Product

A completed **MYP Design ePortfolio covering Criteria C and D**, containing the following deliverables.

Criterion C: Creating the Solution

1. **Strand i – Planning the Solution:** A logical plan showing the sequence of steps required to create the solution.
2. **Strand ii – Technical Skills:** Screenshots or other relevant evidence of technical skills demonstrated during the creation process, accompanied by reflections on the skills used.
3. **Strand iii – Creating the Solution:** A Wokwi circuit simulation, Arduino code, and a reflection explaining the creation process and coding logic.
4. **Strand iv – Changes During Creation:** A record of changes made during creation, the reasons for those changes, and reflections on their impact on the solution.

Criterion D: Evaluating the Solution

1. **Strand i – Evaluation:** Evidence of the completed solution and its final state.
2. **Strand ii – Testing:** Evidence of testing the solution, including relevant screenshots and observations.
3. **Strand iii – Evaluating Success:** An evaluation identifying the strengths and limitations of the solution in relation to its intended purpose and design specifications.
4. **Strand iv – Future Improvements:** A proposed redesign or modification, supported by a sketch or diagram and a reasoned explanation of how it could improve the solution.

The ePortfolio must also include the final reflection and completed submission checklist provided in the template.

Standards

Your work will be assessed against **MYP Design Criterion C: Creating the Solution and Criterion D: Evaluating**, using the achievement-level descriptors and Task-Specific Clarifications (TSC) provided below.

Task-Specific Clarifications (TSC)Criterion C: Creating the Solution

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Demonstrates minimal technical skills when making the solution. ii. Creates the solution, which functions poorly and is presented in an incomplete form.	i. Provides an incomplete or unclear plan for creating the sensor-based system. ii. Provides limited screenshots or evidence of technical skills. iii. Presents an incomplete Wokwi circuit or Arduino code that functions poorly. iv. Records few or no changes made during creation, with little explanation.
3–4	i. Outlines each step in a plan that contains some details, resulting in	i. Outlines the main steps for creating the sensor-based system, but some steps, resources, or sequencing details

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	peers having difficulty following the plan to create the solution. ii. Demonstrates satisfactory technical skills when making the solution. iii. Creates the solution, which partially functions and is adequately presented. iv. Outlines changes made to the chosen design or plan when making the solution.	are unclear. ii. Demonstrates satisfactory technical skills and provides some screenshots of the creation process. iii. Presents a Wokwi circuit and Arduino code that demonstrate some intended functions, although the system may not work consistently. iv. Identifies changes made to the design or plan and outlines basic reasons for them.
5–6	i. Constructs a plan that considers time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates competent technical skills when making the solution. iii. Creates the solution, which functions as intended and is presented appropriately. iv. Outlines changes made to the chosen design and plan when making the solution.	i. Constructs a logical plan that includes the main creation steps, sequence, time, and resources needed to develop the sensor-based system. ii. Demonstrates competent technical skills in Wokwi circuit construction and Arduino coding, supported by relevant screenshots and reflections. iii. Presents a functional simulation with the sensor, Arduino, and output component connected and programmed to perform the intended task. iv. Records changes made during creation and outlines their reasons.
7–8	i. Constructs a logical plan, which outlines the efficient use of time and resources, sufficient for peers to be able to follow to create the solution. ii. Demonstrates excellent technical skills when making the solution. iii. Follows the plan to create the solution, which functions as intended and is presented appropriately. iv. Explains changes made to the chosen design and plan when making the solution.	i. Constructs a detailed, logically sequenced plan that clearly communicates the efficient use of time and resources, enabling a peer to recreate the system. ii. Demonstrates excellent technical skills through accurate circuit construction, effective Arduino coding, debugging, and clear evidence of the creation process. iii. Follows the plan to produce a fully functional Wokwi simulation that accurately implements the chosen design and is clearly presented. iv. Explains changes made to the design or plan and justifies the reasons for those changes in relation to functionality and the intended purpose.

Criterion D: Evaluating

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
0	The student does not reach a standard described by any of the descriptors below.	The student does not provide sufficient evidence to meet any of the task-specific expectations below.
1–2	i. Describes a testing method, which is used to measure the	i. Describes a basic test for the sensor-based system, with limited details about how it will be conducted. ii. States

Grade boundary	IB rubric	Task-Specific Clarification (TSC)
	success of the solution. ii. States the success of the solution.	whether the system works, with little or no supporting evidence from testing or reference to the design specifications.
3–4	i. Describes a relevant testing method, which generates data, to measure the success of the solution. ii. Outlines the success of the solution against the design specification based on relevant product testing. iii. Lists the ways in which the solution could be improved. iv. Outlines the impact of the solution on the client/target audience.	i. Describes a relevant test of the sensor, programmed response, or output and records some observations or results. ii. Outlines whether the system meets some design specifications, using relevant testing evidence. iii. Lists possible improvements to the circuit, code, or system response. iv. Outlines how the proposed system could benefit the intended community or users.
5–6	i. Describes relevant testing methods, which generate data, to measure the success of the solution. ii. Describes the success of the solution against the design specification based on relevant product testing. iii. Outlines how the solution could be improved. iv. Describes the impact of the solution on the client/target audience, with guidance.	i. Describes relevant tests for the sensor-based system and records results that demonstrate how the system responds under the tested conditions. ii. Describes the extent to which the system meets the design specifications, referring to testing results and relevant evidence. iii. Outlines specific improvements to address identified limitations in the circuit, code, or system response. iv. Describes the likely impact of the proposed system on the intended users or community, with guidance.
7–8	i. Describes detailed and relevant testing methods, which generate accurate data, to measure the success of the solution. ii. Explains the success of the solution against the design specification based on authentic product testing. iii. Describes how the solution could be improved. iv. Describes the impact of the solution on the client/target audience.	i. Describes detailed, relevant testing methods and records accurate results across appropriate test conditions to measure the system's performance. ii. Explains the success of the system against the design specifications, explicitly linking authentic simulation or product-testing evidence to each relevant specification. iii. Describes specific, feasible improvements that address identified weaknesses and explains how they could improve functionality. iv. Describes the potential impact of the system on the intended users or community, linking the explanation to the original problem and the solution's demonstrated performance.

Submission Requirements

- Complete all sections of the provided MYP Design ePortfolio.

- Include screenshots of the Wokwi circuit, Arduino code, technical skills, and testing process wherever indicated in the template.
- Ensure that the reflections explain your decisions and are supported by relevant evidence.
- Refer to the design specifications developed in Criteria A and B when evaluating the solution.
- Document changes made during creation, including the reason for each change and its effect on the solution.
- Include testing observations and evidence that demonstrate the performance of the sensor-based system.
- Propose realistic future improvements supported by a redesign sketch or diagram.
- Complete the final reflection and submission checklist before submitting the ePortfolio on ManageBac.

OCT
1

Criteria A and B

Summative Summative 16/18 Students Thursday at 3:00 PM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with some guidance iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution	The student: i. **explains** the community problem clearly on the

Achievement Level	Level Descriptors	Task Specific Clarifications
	to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student:	The student:

Achievement Level	Level Descriptors	Task Specific Clarifications
	i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/ research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. **constructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student: i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible	The student: i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes

Achievement Level	Level Descriptors	Task Specific Clarifications
	design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	so a peer could interpret and understand each concept without further explanation, using clear sketches/ screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

OCT
1

Criteria A and B

Summative Summative 18/20 Students Thursday at 10:05 AM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan

the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with some guidance iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works.

Achievement Level	Level Descriptors	Task Specific Clarifications
		iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/ research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. **constructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student:	The student:

Achievement Level	Level Descriptors	Task Specific Clarifications
	i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/ screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

SEP
30

SA_1

Summative Summative 1/1 Students Wednesday at 2:00 AM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed

or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/ secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds answers with some

Achievement Level	Level Descriptors	Task Specific Clarifications
	some guidance iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. c**onstructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student: i. **develops** a design specification	The student: i. **develops** a precise Design Specifications

Achievement Level	Level Descriptors	Task Specific Clarifications
	which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

SEP
29

Criteria A and B

Summative Summative 18/18 Students Tuesday at 11:35 PM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed or unresolved until someone senses them and responds. Your task is to identify a real community problem, research

it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with some guidance iii. **states** the main features of an	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds

Achievement Level	Level Descriptors	Task Specific Clarifications
	existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/ research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. **constructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student:	The student:

Achievement Level	Level Descriptors	Task Specific Clarifications
	i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/ screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

SEP
29

Criteria A and B

Summative Summative 17/17 Students Tuesday at 11:35 AM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World. Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed

or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with some guidance	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution. ii. **states** 3-4 basic questions across the Research

Achievement Level	Level Descriptors	Task Specific Clarifications
	iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	Plan's People, Planet, and System branches and finds answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/ research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. c**onstructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points. iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.

Achievement Level	Level Descriptors	Task Specific Clarifications
7-8	The student: i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	The student: i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/ screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

SEP
29

Criteria A and B

Summative Summative 16/18 Students Tuesday at 11:35 AM

SA Template

ATL Skills (Category, Cluster, Indicator):

Research Skills – Cluster VI: Information Literacy
Collect, record, and verify data from primary and secondary sources to identify and provide evidence for a real-world problem.

Thinking Skills – Cluster VIII: Critical Thinking
Evaluate evidence, arguments, and existing products to justify a proposed design solution.

Concepts:
Key Concept: Communities
Related Concepts: Function, Innovation

Content/Topic:
Investigating a real community problem and researching, specifying, and planning a sensor-based automated system (sensor → microcontroller → actuator) that solves it within the global context of Scientific and Technical Innovation and ready to prototype in Wokwi.

Statement of Inquiry:
Innovative systems function effectively when engineers use researched, tested evidence to sense a community's needs and design a justified, responsive solution.

Task Details

Situation:
You are a student design engineer completing the IB MYP Design Student Workbook for Sensing Our World.

Communities everywhere face problems : safety risks, environmental hazards, accessibility gaps that go unnoticed or unresolved until someone senses them and responds. Your task is to identify a real community problem, research it thoroughly, analyse existing sensor-based products that address similar problems, and use your findings to design, specify, and plan an automated system, built from sensors (e.g. PIR, rain, gas, temperature, ultrasonic), an Arduino microcontroller, and actuators (e.g. LED, servo motor, buzzer, LCD, relay) that could sense the problem and trigger a response. Your ideas must be tested against clear specifications and justified with a decision matrix before you plan the system architecture and flow diagrams needed to prototype your solution in Wokwi.

Role:
A junior design engineer on a project investigating a real community problem and developing an evidence-based, feasible automated sensing system to solve it.

Audience:
The members of the community affected by your chosen problem (e.g. residents, the school community, or local stakeholders) who would use or benefit from your proposed system, and the client/design panel evaluating whether your solution is well-researched and feasible.

Product:
A completed IB MYP Design Student Workbook: a Problem Statement with context, evidence, and global connections; a prioritized Research Plan (People, Planet, System); an Existing Product Analysis of three or more products; a Research Synthesis and Design Brief; a Design Specifications Document; three annotated Design Ideas; a Decision Matrix with a justified Final Choice; and a System Architecture Diagram and System Flow Diagram, ready to prototype in Wokwi.

Standards:
Your work will be assessed against MYP Design Criterion A (i–iv) and Criterion B (i–iv), detailed in the Task-Specific Clarifications below.

Attaching the ***Task Specific Clarification***

Criterion A: Inquiring and Analysing

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research.	The student: i. **states** the community problem in one short paragraph (e.g. "people forget to bring water bottles"), with minimal to no information on context/effects/global connection, without stating why a solution is needed. ii. **states** two facts about the problem from research (primary/secondary), without connecting them to the stated problem.
3-4	The student: i. **outlines** the need for a solution to a problem ii. **states** some points of research needed to develop a solution, with	The student: i. **outlines** the community problem, briefly noting where it happens or who it affects, along with context and how it connects globally, and provides the need for a solution.

Achievement Level	Level Descriptors	Task Specific Clarifications
	some guidance iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines** some of the main findings of research.	ii. **states** 3-4 basic questions across the Research Plan's People, Planet, and System branches and finds answers with some guidance, without prioritizing the questions. iii. **states** two-three main features of an existing product, without describing how it works. iv. **outlines** some findings from research, without clearly summarizing, leaving gaps or unclear links to the stated problem.
5-6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to develop a solution, with some guidance iii. **outlines** the main features of an existing product that inspires a solution iv. **outlines** the main findings of relevant research.	The student: i. **explains** the community problem clearly on the Problem Statement and "Why Does This Problem Need a Solution?" cards, including at least one consequence of leaving it unsolved. ii. **states and prioritizes** research questions across the People, Planet, and System branches of the Research Plan, with some guidance. iii. **outlines** the main features of at least 3 existing products using an analysis framework; outlines the research synthesis, which does not yet include the features to be adopted, adapted and avoided in their own solution. iv. **outlines** several relevant findings, drawing on both the Research Findings box and the Existing Product Analysis box (at least two products considered), with some explanation of how these findings relate to the needs of the user/client.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed, with minimal guidance iii. **describes** the main features of an existing product that inspires a solution iv. **presents** the main findings of relevant research.	The student: i. **explains and justifies** the community problem in detail across the Problem Statement, Context, and "Why Does This Problem Need a Solution?" cards, using evidence to show why a solution is needed. ii. **states and prioritizes** research questions across all three Research Plan branches independently, with minimal guidance, showing a clear rationale for what to investigate first. iii. **describes** in detail the main features (SWOT) of 3 or more existing products/systems on the Existing Product Analysis; states the research synthesis, which includes the features to be adopted, adapted, and avoided in their own solution. iv. **presents** the main findings clearly and in an organized way across the Research Findings and Existing Product Analysis boxes, explicitly connecting them to the user's needs, and uses these findings to justify the proposed solution concept and the design

Achievement Level	Level Descriptors	Task Specific Clarifications
		brief that follows.

Criterion B: Developing Ideas

Achievement Level	Level Descriptors	Task Specific Clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1-2	The student: i. **lists** a few basic success criteria for the design of a solution ii. **presents** one design idea, which can be interpreted by others iii. **creates** incomplete planning drawings/diagrams.	The student: i. **lists** 1-2 basic points in the Design Specifications Document, with minimal to no connection to the survey/ research data. ii. **presents** only one of the three Design Idea boxes, with a basic sketch or label and little to no annotation. iii. creates the System Architecture and/or Flowchart diagram with missing labels, arrows, or key components (Sensor, Arduino, Output).
3-4	The student: i. **constructs** a list of the success criteria for the design of a solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) or explains key features, which can be interpreted by others iii. **outlines** the main reasons for choosing the design with reference to the design specification iv. **creates** planning drawings/ diagrams or lists requirements for the chosen solution.	The student: i. c**onstructs** a short list of success criteria in the Design Specifications Document, with a loose or unclear link back to the Criterion A research/survey findings. ii. **presents** 2-3 of the Design Idea boxes, with either a sketch or a brief written explanation of the concept, not both. iii. **outlines** an idea in "My Final Choice" and gives 1-2 sentences of reasoning that briefly mentions the specification, without matching each point. iv. **creates** one of the two planning diagrams (System Architecture or Flowchart) accurately, or lists some build requirements without a full diagram.
5-6	The student: i. **develops** design specifications, which identify the success criteria for the design of a solution. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and explains key features, which can be interpreted by others. iii. **presents** the chosen design and outlines the main reasons for	The student: i. **develops** a full Design Specifications Document with clear, identifiable success criteria drawn from the Criterion A research/survey data. ii. **presents** all three Design Idea boxes, each with an annotated sketch/screenshot and a written explanation of the concept's key features. iii. **presents** a decision matrix to compare the three ideas, then outlines the main reasons for the final choice in "My Final Choice," referencing specific specification points.

Achievement Level	Level Descriptors	Task Specific Clarifications
	its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and lists requirements for the creation of the chosen solution.	iv. **creates** both the System Architecture diagram (Sensor → Arduino → Output) and the Flowchart accurately, with labelled components and a basic list of requirements for the build.
7-8	The student: i. **develops** a design specification which outlines the success criteria for the design of a solution based on the data collected. ii. **presents** a range of feasible design ideas, using an appropriate medium(s) and annotation, which can be correctly interpreted by others. iii. **presents** the chosen design and outlines the reasons for its selection with reference to the design specification. iv. **develops** accurate planning drawings/diagrams and outlines requirements for the creation of the chosen solution.	The student: i. **develops** a precise Design Specifications Document where each success criterion is explicitly traceable to a specific piece of Criterion A data (e.g. survey figures, source findings). ii. **presents** fully annotates all three Design Idea boxes so a peer could interpret and understand each concept without further explanation, using clear sketches/ screenshots and detailed feature notes. iii. **presents** the final choice with a detailed, point-by-point justification in "My Final Choice," explicitly linking each reason back to the decision matrix and design specification. iv. **develops** accurate, fully labelled System Architecture and Flowchart diagrams and outlines the complete sequence of requirements needed to build the prototype.

MYP Assessment Criteria

6/8 ☑ A: Inquiring and analysing

6/8 ☑ B: Developing ideas

N/A ☑ C: Creating the solution

N/A ☑ D: Evaluating

Description

📖 Learning Experiences

Prior Learning Experiences

The prior learning experiences of students that relate to this unit

1. Basic Scientific Concepts:

- Students might have learned about basic concepts in physics and energy, which can be connected to the functioning of sensors and their interactions with the environment.

2. Technology Interaction:

- Students may have used smartphones, touchscreens, and other devices with sensors, giving them a basic understanding of how technology responds to inputs.

3. Environmental Studies:

- Students may have studied topics related to environmental sustainability, pollution, and the impact of human activities on the environment.

4. Mathematical Concepts:

- Students may have learned about units of measurement, which can be applied when discussing sensor readings.

Assessment of prior learning will be achieved by,

1. Concept Recall Quiz

2. Technology Interaction Discussion

3. Environmental Impact Brainstorm

Reflections

General Reflections

Prior to studying the unit



Chaitra R Aug 18, 2023 at 11:25 AM

This unit has been thoughtfully designed to provide students with a unique opportunity to apply their electronics knowledge in a hands-on and practical manner. By creating devices that can interact with the surrounding environment using various sensors, students will not only deepen their understanding of electronics but also contribute to the exploration and understanding of their surroundings.

The selection of topics in the "Sensing the Environment" unit is inherently captivating as it bridges the gap between theoretical knowledge and real-world application. By creating devices that can sense and respond to environmental factors, students will experience the direct relevance of electronics in their everyday lives.

Students already possess a foundational understanding of electronics principles, circuitry, and components from their previous studies. This knowledge equips them with the basics needed to delve deeper into the integration of sensors and electronic components to create responsive devices.

The "Sensing the Environment" unit provides ample opportunities for the development of key attributes within the learner profile. Problem-solving skills will be nurtured as students troubleshoot and optimize their sensor-integrated devices. Inquisitiveness will be encouraged as students explore the different types of environmental sensors and their applications. Collaborative skills will be enhanced through group work and peer feedback during the design and creation process.

Lessons & Resources

Core Framework of Smart Systems: Sensor --> Processor --> Actuator

Description & Guidance

In today's session, students revisited the foundational **Sensor --> Processor --> Actuator** framework that underpins automated technology.

They explored how various sensors perform distinct functions, drawing parallels to how human sense organs possess specific sensory capabilities. Key components discussed during the session included:

- **Sensors:** IR, Ultrasonic, PIR, Temperature, and Gas sensors.
- **Actuators:** LEDs, motors, and buzzers.
- **Processors:** Microcontrollers such as Arduino Uno and Nano.

To ground these concepts in real-world applications, the teacher demonstrated how the **Sensor --> Processor --> Actuator** model functions within automated sliding doors and rainwater level alert systems.

Please refer to the attached presentation slide for visual reference. Students concluded the session by completing a review quiz to consolidate their learning and attaching it here.

sensing-our-world-quiz.html
456.74 KB

Teacher Notes

In today's session, students revisited the foundational **Sensor --> Processor --> Actuator** framework that underpins automated technology.

They explored how various sensors perform distinct functions, drawing parallels to how human sense organs possess specific sensory capabilities. Key components discussed during the session included:

- **Sensors:** IR, Ultrasonic, PIR, Temperature, and Gas sensors.
- **Actuators:** LEDs, motors, and buzzers.
- **Processors:** Microcontrollers such as Arduino Uno and Nano.

To ground these concepts in real-world applications, the teacher demonstrated how the **Sensor --> Processor --> Actuator** model functions within automated sliding doors and rainwater level alert systems.

1

Ishita Bhattacharjee
Posted **task** on Aug 6, 2026 at 1:19 PM

AUG 20

Quiz: Sensors and Intelligent systems
Classwork Thursday at 9:30 AM

In this task, you will complete an **onlin**

sensing-our-world-quiz.html
456.74 KB

e quiz to demonstrate your understanding of the concepts covered in recent lessons. The quiz will assess your knowledge of **machines, automated systems, sensors, and different types of sensors**, including their purpose,

functions, and real-world applications. Read each question carefully and answer to the best of your ability. This quiz will help you reflect on your learning and identify areas for further improvement.

Slide used in class

PEEL for writing a draft_Scaffold for Criteria A

Description & Guidance

In this session, students learnt how the PEEL framework (Point, Evidence, Explanation, and Link) can be used to justify the need for a design solution through clear and structured analytical writing. They activated prior knowledge from MYP 1 Humanities, demonstrating ATL transfer skills by applying an existing writing strategy in a new subject context. Through class discussion, students explored the importance of using the CRAAP test to evaluate the credibility of research evidence and incorporate it effectively into the PEEL structure. They then practised applying the framework by completing the "PEEL Practice Task," using primary and secondary research to analyse a design problem and justify the need for a solution.

PEEL.png
1.59 MB

Teacher Notes

In this session, students learnt how the PEEL framework (Point, Evidence, Explanation, and Link) can be used to justify the need for a design solution through clear and structured analytical writing. They activated prior knowledge from MYP 1 Humanities, demonstrating ATL transfer skills by applying an existing writing strategy in a new subject context. Through class discussion, students explored the importance of using the CRAAP test to evaluate the credibility of research evidence and incorporate it effectively into the PEEL structure. They then practised applying the framework by completing the "PEEL Practice Task," using primary and secondary research to analyse a design problem and justify the need for a solution.

1

Ishita Bhattacharjee
Posted **task** on Aug 6, 2026 at 9:42 AM

AUG 11

PEEL Practice Task
Classwork 18/18 Students Tuesday at 10:30 AM

Scenario

Your design team has been asked to investigate why **single-use plastic water bottles continue to be widely used in schools despite reusable bottles being available.**

The school would like to design a product, campaign, or system that encourages students to make more responsible choices. Before developing a solution, you must first understand and analyse the problem using research.

Primary Research

A survey was conducted among **120 students**.

Survey Results

- **35%** bring a reusable water bottle to school every day.
- **58%** buy bottled water from the canteen at least twice a week.
- **62%** admit they often forget to carry a reusable bottle.
- **54%** believe buying bottled water is more convenient.
- **38%** think that their individual actions have little impact on the environment.
- **79%** said they would use reusable bottles more often if refill stations were easier to access.

Interview Responses

Student A

"I know plastic bottles are bad for the environment, but I usually forget my reusable bottle."

Student B

"Buying a bottle is faster and more convenient than carrying one."

Student C

"If there were more refill stations around the school, I would definitely use my reusable bottle."

Secondary Research

Research from environmental organisations reports that:

- Approximately **1 million plastic bottles are purchased every minute worldwide.**
- Many plastic bottles are **not recycled** and can remain in the environment for **hundreds of years.**
- Plastic pollution harms wildlife, especially marine animals, through ingestion and entanglement.
- Manufacturing plastic bottles requires large amounts of petroleum, water, and energy, increasing greenhouse gas emissions.
- Studies show that **convenience and habit** are among the biggest reasons why consumers continue to choose single-use plastics.

Your Task

Using the information provided, write **one PEEL paragraph (180–250 words)** answering the following question.

How does irresponsible consumer behaviour contribute to the excessive use of single-use plastic bottles, and why is this an important design problem?

Writing Guide

Point

State the design problem in one clear sentence.

Sentence starter:

One major design problem is that...

Evidence

Include:

- At least **two pieces of evidence** from the primary research.
- At least **one piece of evidence** from the secondary research.

Sentence starters:

- **According to the survey...**
- **Interview responses showed...**
- **Secondary research indicates...**

Explanation

Explain:

- Why this behaviour occurs.

- The causes of the problem.
- The effects on people and the environment.
- How the evidence supports your point.

Sentence starters:

- **This suggests that...**
- **One reason for this behaviour is...**
- **As a result...**
- **This demonstrates that...**

Link

Conclude by explaining why this problem matters.

Include:

- **Local impact** (school or community)
- **Global impact** (environment, climate, wildlife, waste)

Sentence starters:

- **Therefore...**
- **This problem affects our school because...**
- **On a global scale...**

Success Checklist

☐ I identified a clear design problem.

☐ I used evidence from the primary research.

☐ I used evidence from the secondary research.

☐ I explained the causes of the problem.

☐ I explained the effects of the problem.

☐ I linked the problem to both local and global impacts.

☐ I followed the PEEL structure throughout my paragraph.

Investigating the Problem & Justifying the Need (Criterion A.i)

Description & Guidance

- **What we did today:** Students worked on their Formative Assessment (FA 1) for MYP 2 Design. They analyzed survey results and secondary research about middle school hydration habits to practice explaining a design problem and justifying why a new product or solution is needed.
- **Key Skills Practiced:**
  - **Critical Thinking:** Synthesizing primary survey data and secondary research facts.
  - **Structured Writing:** Using the **PEEL format** (Point, Evidence, Explanation, Link) to build clear, persuasive arguments without copying source text directly.
  - **Self-Assessment:** Evaluating their own work against the MYP 2 Task-Specific Clarification rubric.
- **How you can support at home:** Ask your child to explain their **PEEL** paragraphs to you or share one key piece of evidence

they used from their research to justify why students need better hydration solutions at school.

Teacher Notes

In today's MYP 2 Design class, students focused on unpacking the inquiry process by identifying a real-world problem and justifying the need for a solution. Using a given scenario regarding student hydration habits, primary survey data, and secondary research facts, students applied the **PEEL (Point, Evidence, Explanation, Link)** framework to structure two well-developed analytical paragraphs. We reviewed the Task-Specific Criteria (TSC) using a developmental scale (**Proficient**, **Developing**, **Emergent**) to help students self-assess their ability to explain the problem in their own words, integrate research evidence, and justify why a design solution is required.

1

Ishita Bhattacharjee
Posted **task** on Aug 6, 2026 at 12:08 PM

AUG 11

FA 1 : PEEL _Problem Statement and Justifying the Need for a Solution

Formative Formative 18/18 Students Tuesday at 11:35 AM

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10

Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

Task Specific Clarification

Performance Level	Rubric Descriptors
Proficient	• Explains clearly the problem faced by students in their own words • Integrates evidence effectively from both primary and secondary research • Justifies convincingly why a new design solution/product is needed • Structure: Follows the PEEL structure consistently across both paragraphs.
Developing	• Outlines or states the problem, though description may closely paraphrase sources • Includes basic evidence from research, but connections are brief • Attempts to justify the need for a solution with basic reasoning • Structure: Follows most aspects of the PEEL structure.
Emergent	• States a problem, but it lacks focus or accuracy • Uses minimal or no evidence from the research • Offers limited justification for why a solution is required • Structure: Does not follow the PEEL framework consistently or copies text directly.

Resources

Devendhar (Dev) Bachhu
Posted **website** on Oct 29, 2026 at 9:27 AM

Task Sheet

https://docs.google.com/spreadsheets/d/1ZLDoa4OZJj1z5VakPijdprwZQD0lh7q9C06E8qosxB8/edit?gid=0#gid=0

Devendhar (Dev) Bachhu
Posted **website** on Sep 30, 2026 at 9:22 AM

Quiz Link

https://devs67.github.io/MYP-Design-Digital/myp1-2/criterion-b-fa-myp2.html?theme=dark#q=m2&s=9u8h9x&a=-.-.-.-----.-.-.-.----.-.-.-.-.-----.----.-.-.-.-.-.-.-.-.------.-.-.-.-&n=&c=

Devendhar (Dev) Bachhu
Posted **website** on Sep 25, 2026 at 6:12 PM

Simulation Platform

https://www.circuito.io/

Devendhar (Dev) Bachhu
Posted **website** on Sep 25, 2026 at 3:15 PM

Isometric Grid

https://www.teacherled.com/iresources/shapesapp/isometricdots/isometricdots.html

---

Devendhar (Dev) Bachhu
Posted **website** on Sep 25, 2026 at 2:43 PM

Videos For Isometric Drawings

https://drive.google.com/drive/folders/1eVjUq2zmwfVYfpvDn1v3K9neq1al1gF6?usp=drive_link

---

Devendhar (Dev) Bachhu
Posted **website** on Sep 17, 2026 at 10:10 AM

Crit B-i- PPT

https://docs.google.com/presentation/d/1rTLNH1OPsKOWqsuyQb4VmKLf_9ZArzpZxLi6qrs0c38/edit?usp=sharing

---

Devendhar (Dev) Bachhu
Posted **website** on Sep 17, 2026 at 9:09 AM

Crit - A - iii- PPT

https://docs.google.com/presentation/d/1cmkAMoP3hZTt7UuMA4QYQoII8zhAL3j8jLdWA09wzdU/edit?usp=sharing

---

Devendhar (Dev) Bachhu
Posted **website** on Sep 9, 2026 at 12:46 PM

PPT

Strand-iii

https://docs.google.com/presentation/d/1cmkAMoP3hZTt7UuMA4QYQoII8zhAL3j8jLdWA09wzdU/edit?usp=sharing

---

Sushma Goyal
Posted **website** on Aug 27, 2026 at 11:04 AM

Types of sensors

https://www.geeksforgeeks.org/electronics-engineering/types-of-sensors/

Sushma Goyal
Posted **1 file** on Aug 27, 2026 at 11:00 AM

Sensors and Different components

Sensing_Our_World__Automated_Systems_for_Communities.pdf
3 MB PDF Document

Moin Chorgay
Posted **website** on Jan 19, 2024 at 11:41 AM

MYP Design Cycle

https://sites.google.com/mymail.lausd.net/mypdesigncycle/home?authuser=0